$\text{Vol(standard simplex in } \mathbb{R}^d) = \frac{1}{d!}$

if p = 1, then V(1)

$V_d^p(r) = V_d^p(1) \cdot r^d$ scaling all dimensions

$V_d^p(1) = 2^d \int \cdots \int_{x_1^p + \cdots + x_d^p \le 1,\ x_i \ge 0} dx_1 \cdots dx_d$ instead of all sum (absd)<1, we take pos. orthan and sc.

Exercises: Phenomena in High Dimensions Mathematical

Foundations for Machine Learning

gamma function = factorials for any number -> smooth with recursive restriction
beta function = related to gamma (x, y), geometrically is the integral of a curve between two points (related to the powers of x and y)

Volume of ℓ_p-Balls and the Gamma Function

Exercise 1.1 [Easy]

The volume of the unit ℓ_p-ball in $\mathbb{R}^d$ is given by

$$V_d^p(1) = \frac{\left(2\,\Gamma\left(\frac{1}{p}+1\right)\right)^d}{\Gamma\left(\frac{d}{p}+1\right)}.$$

from Lp norm: $|x_2|^p + \cdots + |x_d|^p \le 1 - |x_1|^p$

(d-1) dim ball with R radius $R = (1 - |x_1|^p)^{1/p}$

pick x1 and integrate over d-1 to get d volume

$V_d^p(1) = \int_{-1}^{1} V_{d-1}^p\left((1-|x_1|^p)^{1/p}\right) dx_1$

3d ball is sum of all 2d circles -> integrate

(a) Using the identity $\Gamma(x+1) = x\,\Gamma(x)$ and $\Gamma(1) = 1$, compute $\Gamma(4)$.

(b) Using $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$, compute $\Gamma\left(\frac{5}{2}\right)$.

(c) Compute the exact volume of the unit ℓ_2-ball in $\mathbb{R}^4$.

$V_d^p(1) = V_{d-1}^p(1) \int_{-1}^{1} (1-|x_1|^p)^{\frac{d-1}{p}}\, dx_1$

apply symm.

$V_d^p(1) = 2\,V_{d-1}^p(1) \int_{0}^{1} (1-x_1^p)^{\frac{d-1}{p}}\, dx_1$

gamma(n) = (n-1)!

change of var.: x1=z1/p

$V_d^p(1) = \frac{2\,V_{d-1}^p(1)}{p} \int_0^1 z^{\frac{1}{p}-1}(1-z)^{\frac{d-1}{p}}\, dz$

Exercise 1.2 [Easy]

as we consider bigger area the prop. inside the ball is less (how many d dim cubes)

Recall that the volume of the unit ℓ_1-ball in $\mathbb{R}^d$ is $V_d^1(1) = \frac{2^d}{d!}$.

recognize beta

$V_d^p(1) = \frac{2\,V_{d-1}^p(1)}{p} \cdot B\left(\frac{1}{p}, \frac{d-1}{p}+1\right)$

(a) Compute $V_d^1(1)$ for $d = 2, 4, 6, 10$ and describe the qualitative behavior.

(b) The volume of the unit ℓ_∞-ball in $\mathbb{R}^d$ is $V_d^\infty(1) = 2^d$. Compare the grow

$V_d^p(1) = 2\,V_{d-1}^p(1) \cdot \frac{\Gamma\left(\frac{1}{p}+1\right)\Gamma\left(\frac{d-1}{p}+1\right)}{\Gamma\left(\frac{d}{p}+1\right)}$

$V_d^1(1)$, $V_d^2(1)$, and $V_d^\infty(1)$ as $d \to \infty$. Which of these converge to zero?

absorb 1/p in gamma

Exercise 1.3 [Medium]

The volume of the ℓ_p-ball of radius r in $\mathbb{R}^d$ scales as $V_d^p(r) = V_d^p(1)\, r^d$.

$V_d^p(1) = V_1^p(1) \cdot \prod_{l=2}^{d} C(l) = V_1^p(1) \cdot \prod_{l=2}^{d} 2 \cdot \frac{\Gamma\left(\frac{1}{p}+1\right) \cdot \Gamma\left(\frac{l-1}{p}+1\right)}{\Gamma\left(\frac{l}{p}+1\right)}$

(a) Show that the radius r_d such that $V_d^1(r_d) = 1$ satisfies

apply recursion

$$r_d = \frac{(d!)^{1/d}}{2}.$$

$V_d^p(1) = 2$ bec. [-1, 1] one dim

(b) Using Stirling's approximation $n! \approx \sqrt{2\pi n}\,(n/e)^n$, show that for large d,

$V_d^p(1) = 2 \cdot \left(2\,\Gamma\left(\frac{1}{p}+1\right)\right)^{d-1} \cdot \frac{\Gamma\left(\frac{1}{p}+1\right)}{\Gamma\left(\frac{d}{p}+1\right)}$

radius grow to mantain volume with dimension growth is linear

$$r_d \approx \frac{d}{2e}.$$

- take out constant from product
- telescope division

(c) Similarly, for the ℓ_2-ball, one can show that the unit-volume radius scales as $r_d \approx \sqrt{d/e}$ for even dimensions. Interpret why $r_d \to \infty$ for both ℓ_1 and ℓ_2, and explain why this phenomenon does *not* occur for the ℓ_∞-ball.

You don't need to compensate in infinity norm (shape of norm)

Exercise 1.4 [Medium]

Let $A_d^2(r)$ denote the surface area of the ℓ_2-ball (sphere) of radius r in $\mathbb{R}^d$. It is related to the volume by

$$A_d^2(r) = \frac{d}{dr} V_d^2(r).$$

(a) Using $V_d^2(r) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}\, r^d$, derive a formula for $A_d^2(r)$.

(b) Compute the surface area of the unit sphere in $\mathbb{R}^3$ and $\mathbb{R}^4$.

(c) Show that $\lim_{d\to\infty} A_d^2(1) = 0$. What does this imply about the geometry of high-dimensional spheres?

if you take inner ball and d -> infinity, inner ball's volume goes faster to 0, then what remains is surface

$\frac{dV_d^2(R)}{dR} = V_d^2(1) \cdot d \cdot R^{d-1}$

Exercise 1.5 [Hard]

We know $V_d^2(1) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$ and $V_d^1(1) = \frac{2^d}{d!}$.

area in multidimension = derivate of volume.
in 3d ball, the area is the surface of the ball, so when increasing a bit area then volume increases

$r = \sqrt{1-a^2}$ $V_{d-1}^2(1) \cdot (1-a^2)^{(d-1)/2}$

(a) Show that for even dimensions $d = 2k$,

$$\frac{V_d^1(1)}{V_d^2(1)} = \left(\frac{2}{\sqrt{\pi}}\right)^{2k} \frac{1}{\prod_{\ell=k+1}^{2k} \ell}.$$

when we take a = x^2 in -1, 1 , substitute the r, then the volume is concentrated where a = 0. In 3D ball the max vol will be in the equator, the biggest circle slice. In multidim taking power of d will be more concentrated in the eq.

we take V1 / V2 and show that when d -> infinity then L1 goes to 0 first -> the l1 ball is much smaller

Hint: Express both volumes in terms of factorials and simplify.

(b) Using Stirling's formula or a direct bound, show that $\frac{V_d^1(1)}{V_d^2(1)} \to 0$ as $d \to \infty$, i.e., the ℓ_1-ball volume decays *strictly faster* than the ℓ_2-ball volume.

(c) Give a geometric interpretation: why should the ℓ_1-ball be "much smaller" than the ℓ_2-ball in high dimensions, despite the containment $B_1(0,1) \subseteq B_2(0,\sqrt{d})$?

When you take d to infinity, ||X|| by law of large numbers converges to the E[X] which is root(d), all same distance from theorigin, almo

Concentration of Volume Under the Surface

Exercise 2.1 [Easy]

Let $B_d = B_d(0,1)$ denote the unit ℓ_2-ball in $\mathbb{R}^d$ and $B_d(0,1-\varepsilon)$ the ball of radius $1-\varepsilon$ for $0 < \varepsilon < 1$.

(a) Compute the ratio $\frac{V_d^2(1-\varepsilon)}{V_d^2(1)}$ and simplify.

(b) Evaluate this ratio for $\varepsilon = 0.05$ and $d = 10, 50, 100, 500$.

(c) In words, what happens to the fraction of volume in the thin outer shell $B_d(0,1) \setminus B_d(0,1-\varepsilon)$ as $d \to \infty$?

Exercise 2.2 [Medium]

Let $B_d^p(0,r)$ denote the ℓ_p-ball of radius r in $\mathbb{R}^d$.

(a) Show that for any $p \geq 1$ and any $0 < \varepsilon < 1$,

$$\frac{\mathrm{vol}\big(B_d^p(0,1-\varepsilon)\big)}{\mathrm{vol}\big(B_d^p(0,1)\big)} = (1-\varepsilon)^d.$$

Hint: Use the scaling $V_d^p(r) = V_d^p(1)\, r^d$.

(b) Fix $\varepsilon = 0.01$. Find the smallest dimension d_0 such that at least 99% of the volume of the unit ℓ_p-ball lies in the outer shell $B_d^p(0,1) \setminus B_d^p(0,1-\varepsilon)$.

(c) The result in (a) is independent of p. Give an intuitive explanation for why the concentration of volume near the surface is universal across all ℓ_p-balls.

Exercise 2.3 [Hard]

Consider the unit ℓ_2-ball $B_d(0,1)$ in $\mathbb{R}^d$ and define the "equatorial slab" of thickness $\delta > 0$ as

$$S_\delta = \big\{x \in B_d(0,1) \;\big|\; |x_1| \leq \delta\big\}.$$

The volume of S_δ relative to $B_d(0,1)$ can be expressed as

$$\frac{\mathrm{vol}(S_\delta)}{\mathrm{vol}(B_d(0,1))} = \frac{\displaystyle\int_{-\delta}^{\delta} (1-t^2)^{(d-1)/2}\, dt}{\displaystyle\int_{-1}^{1} (1-t^2)^{(d-1)/2}\, dt}.$$

(a) Explain why the integrand $(1-t^2)^{(d-1)/2}$ for the cross-sectional volume appears in this formula.

(b) Show that for $|t| \leq 1$, one has $(1-t^2)^{(d-1)/2} \leq e^{-(d-1)t^2/2}$.

Hint: Use $\ln(1-u) \le -u$ for $u \in [0,1)$.

(c) Using the bound from (b), argue that for any $\delta > 0$ fixed, the fraction of volume *outside* the slab S_δ goes to zero as $d \to \infty$. That is, almost all the volume concentrates near the equator $x_1 = 0$.

Stirling's Formula and Asymptotic Scaling

Exercise 3.1 [Easy]

Stirling's approximation states that for large n,

$$n! \approx \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

(a) Compute 10! exactly and via Stirling's approximation. What is the relative error?

(b) Using Stirling's formula, show that $\ln(n!) \approx n\ln n - n + \frac{1}{2}\ln(2\pi n)$ for large n. Which term dominates?

Exercise 3.2 [Medium]

The volume of the unit ℓ_2-ball in $\mathbb{R}^d$ is $V_d^2(1) = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$.

(a) For even $d = 2k$, we have $\Gamma(k+1) = k!$. Use Stirling's formula to show that

$$\ln V_{2k}^2(1) \approx k\ln\pi - k\ln k + k - \frac{1}{2}\ln(2\pi k)$$

for large k.

(b) Show that the dominant term in $\ln V_{2k}^2(1)$ for large k is $k\ln(\pi/k)$, which is negative for $k \ge 4$. Conclude that $V_d^2(1) \to 0$ *exponentially fast* in d.

(c) Determine the dimension d^* (approximately) at which $V_d^2(1)$ is maximized.

Hint: For even $d = 2k$, find the k where $\ln V_{2k}^2(1)$ transitions from increasing to decreasing.

Concentration of Gaussian Measure

Exercise 4.1 [Easy]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$, so that $X = (X_1, \dots, X_d)$ with independent components $X_i \sim \mathcal{N}(0, \sigma^2)$.

(a) Compute $\mathbb{E}[\|X\|^2]$ and $\mathrm{Var}(\|X\|^2)$.

Hint: $\|X\|^2 = \sum_{i=1}^d X_i^2$, each X_i^2 has mean σ^2 and variance $2\sigma^4$.

(b) Use Chebyshev's inequality to show that

$$P\big(\big|\,\|X\|^2 - d\sigma^2\big| \ge t\big) \le \frac{2d\sigma^4}{t^2}.$$

Exercise 4.2 [Medium]

Let $X \sim \mathcal{N}(0, \sigma^2 I_d)$.

(a) Using the Chebyshev bound from the previous exercise, show that

$$P\Big(\big|\,\|X\|^2 - d\sigma^2\big| \ge t\sqrt{d}\,\sigma^2\Big) \le \frac{2}{t^2}.$$

(b) Re-interpret this as: the squared norm $\|X\|^2$ is within a factor of $1 \pm \frac{t}{\sqrt{d}}$ of $d\sigma^2$ with probability at least $1 - \frac{2}{t^2}$.

(c) For $d = 1000$ and $\sigma = 1$, find the interval $[a, b]$ such that $\|X\|^2 \in [a, b]$ with probability at least 0.95.

Exercise 4.3 [Hard]

In machine learning, we often compute pairwise distances between data points. Consider two independent samples $X, Y \sim \mathcal{N}(0, \sigma^2 I_d)$ in $\mathbb{R}^d$.

(a) Compute $\mathbb{E}[\|X - Y\|^2]$ and $\mathrm{Var}(\|X - Y\|^2)$.

(b) Using Chebyshev's inequality, show that

$$P\left(\left|\frac{\|X - Y\|^2}{2d\sigma^2} - 1\right| \geq \varepsilon\right) \leq \frac{2}{d\varepsilon^2}.$$

(c) Similarly show that $\|X\|^2/(d\sigma^2) \to 1$ in probability. Conclude that for large d, the ratio $\|X - Y\|^2/\|X\|^2$ concentrates around a constant. What is this constant, and what does it imply for the usefulness of Euclidean distance in high-dimensional machine learning?

Mixed Challenging Exercises

Exercise 5.1 [Hard]

Let $X \sim \mathrm{Uniform}(B_d^2(0, 1))$, i.e., X is drawn uniformly from the unit ℓ_2-ball in $\mathbb{R}^d$.

(a) Using the concentration of volume near the surface, argue that $\|X\|$ is close to 1 with high probability as $d \to \infty$. Specifically, show that for any $\varepsilon > 0$,

$$P(\|X\| \leq 1 - \varepsilon) = (1 - \varepsilon)^d \to 0.$$

(b) Now consider $X \sim \mathcal{N}(0, I_d)$. From concentration results, we know $\|X\| \approx \sqrt{d}$ with high probability. If we project onto the unit sphere, $\hat{X} = X/\|X\|$, argue that the first coordinate $\hat{X}_1$ concentrates around 0 as $d \to \infty$.

Hint: $\hat{X}_1 = X_1/\|X\|$ and $X_1 \sim \mathcal{N}(0, 1)$.

(c) Combine (a) and (b) with the equatorial slab result to give a coherent picture: in high dimensions, random points in the ball are near the surface, and points on the surface are near the equator (relative to any fixed direction). What does this imply about the distribution of angles between random vectors?

Exercise 5.2 [Hard]

Consider a machine learning setting where you generate N data points $x_1, \ldots, x_N \sim \mathcal{N}(0, I_d)$ and classify them using a nearest-neighbor classifier.

(a) Using the result that $\|x_i - x_j\|^2/(2d) \xrightarrow{P} 1$ for all $i \neq j$, show that the relative contrast

$$\rho_d = \frac{d_{\max} - d_{\min}}{d_{\min}}, \quad \text{where } d_{\max} = \max_j \|x_1 - x_j\|,\ d_{\min} = \min_{j \neq 1} \|x_1 - x_j\|,$$

satisfies $\rho_d \xrightarrow{P} 0$ as $d \to \infty$.

(b) Suppose the data points lie on a k-dimensional linear subspace of $\mathbb{R}^d$ (with $k \ll d$). Specifically, $x_i = Az_i$ where $A \in \mathbb{R}^{d \times k}$ with orthonormal columns and $z_i \sim \mathcal{N}(0, I_k)$. Compute $\mathbb{E}[\|x_i - x_j\|^2]$ and explain why the concentration of distances now depends on k rather than d.

(c) This is the motivation behind dimensionality reduction techniques (PCA, random projections) used before applying k-NN classifiers. Explain in 2–3 sentences how the analysis from (a) and (b) justifies this preprocessing step.